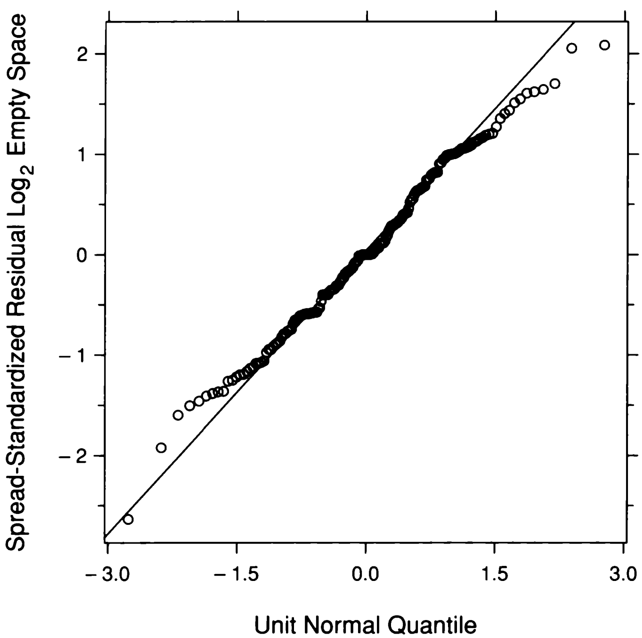
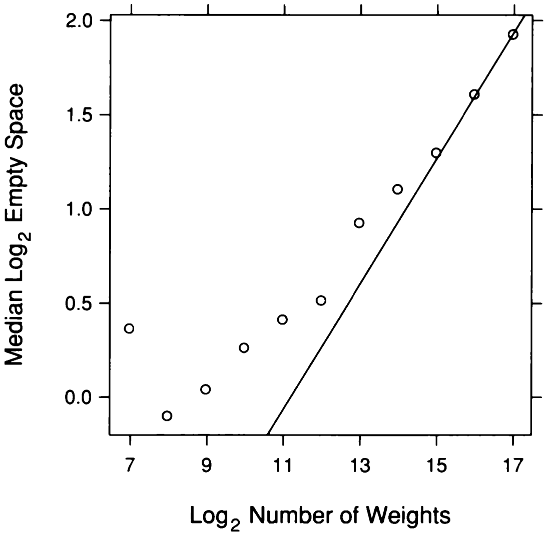
While Figure 2.46 has suggested some differences among the seven distributions with $n > 1000$, the magnitudes of the differences appear small, so we will pool the residuals anyway to further study the platykurtosis of the distributions. Figure 2.47, a normal q-q plot of the pooled values, shows that the shortening of the tails with respect to the normal begins near the quartiles of the data.



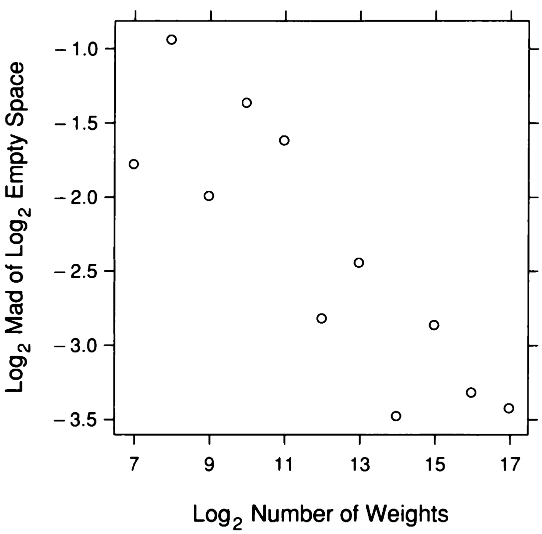
2.47 The normal q-q plot compares the normal distribution with the distribution of the pooled spread-standardized residuals for seven values of n.

Figure 2.48 graphs ℓ_n against $\log n$. Theoretical results suggest that when n gets large, ℓ_n is linear in $\log n$ with a slope of $1/3$ [8]. The line in Figure 2.48 has as slope of $1/3$ and passes through the rightmost point. The points for the largest values of $\log n$ do indeed appear to be increasing with this slope, but for the smaller values of $\log n$, the pattern of the points is convex and the slopes are less. In other words, before the asymptotics take over, the rate of growth in log empty space is less than $1/3$, and the rate increases as n increases.



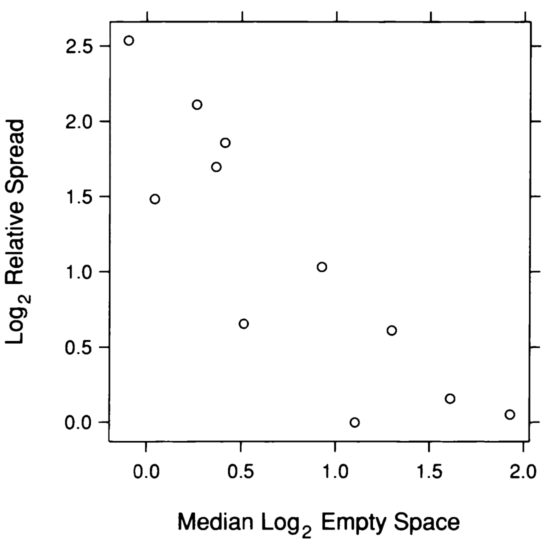
2.48 Median log empty space is graphed against log number of weights.

To study the behavior of s_n as a function of n, Figure 2.49 graphs log s_n against log n. The underlying pattern is linear, and a line with a slope of $-1/3$ would fit the pattern. This triggers an uncomfortable thought.



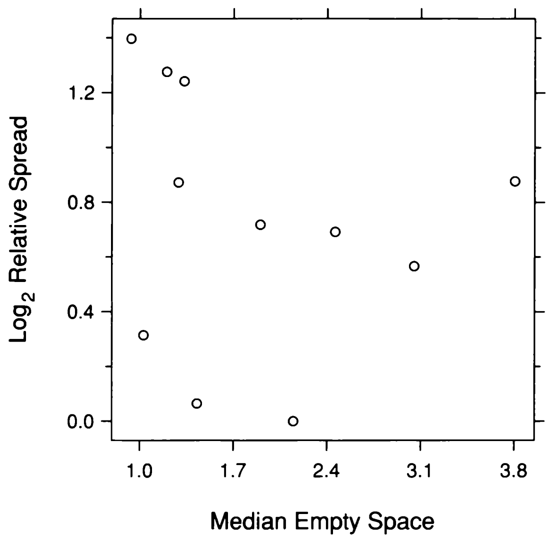
2.49 The logs of the mads for log empty space are graphed against log number of weights.

Since the log medians grow with a slope of $1/3$ as a function of log n, and the log mads grow with a slope of $-1/3$ as a function of log n, the log mads as a function of the log medians should have a slope of -1. This is checked in Figure 2.50; each s_n is divided by the minimum of the s_n, the log is taken, and the values are graphed against ℓ_n. This is simply an alternative form of the s-l plot; instead of graphing square root absolute residuals and square root mads against the medians, the log relative mads are graphed against the medians. The pattern is indeed linear with a slope of -1. In other words, there is monotone spread — a decrease in the spread with location. The discomfort is this. If the spreads of distributions do not depend on the locations, then taking logs can create monotone spread with exactly the pattern observed in Figure 2.50. We took the logs of the measurements of empty space at the outset to enhance the interpretation of multiplicative effects, but it is now likely that the transformation has induced the monotone spread.



2.50 Log relative mads for log empty space are graphed against median log empty space.

This is checked in Figure 2.51; the alternate s-l visualization method employed in Figure 2.50 is also used, but for empty space without transformation. Quite clearly, there is no monotone spread.



2.51 Log relative mads for empty space are graphed against median empty space.

Backing Up

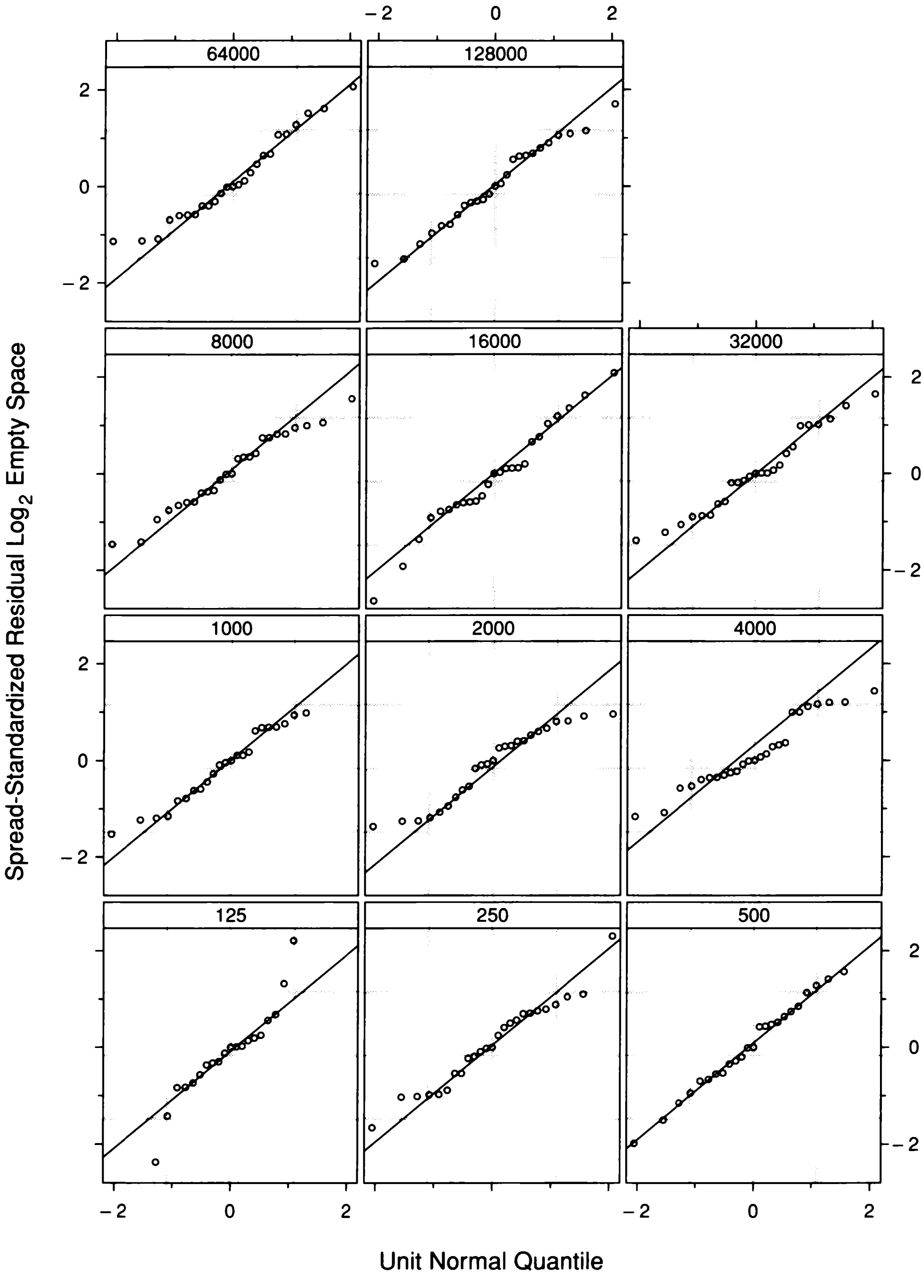
By taking logs at the outset to enhance the interpretation of multiplicative effects, we have induced some of the variation in the spreads of the distributions of empty space. With the benefit of visualization hindsight, it appears that we would be better off fitting empty space rathei than log empty space. Then, we can always take logs of fits to study multiplicative properties. Thus, we should go back to the beginning and analyze the data again without transformation. We will not do so in this account since the data have been treated enough. Such backing up is common. It speaks to the power of visualization to show us mistaken actions that we have no way of judging until we visualize the results.

2.9 Direct Manipulation: Outlier Deletion

Outliers not only force us to use robust fitting, they ruin the resolution of graphical displays. A single large outlier that expands the range of the data by a factor of, say 2, squashes all of the remaining data into 50% of the scale on a graph. The normal q-q plot of the spread-standardized residuals in Figure 2.45 was the victim of such squashing. The range of all spread-standardized residuals is -3.59 to 7.17. But 97% of them lie between ± 2.75, so 3% of the spread-standardized residuals take up about 50% of the vertical scale. The result is a reduction in our ability to assess the nonnormality of the distributions aside from the outlier behavior. For example, the platykurtosis for n greater than 1000 is barely noticeable on the display. The remedy was to continue the visualization for the distributions with $n > 1000$. This removed the outliers. Another sensible procedure would have been to make Figure 2.45 again, but deleting the points whose vertical scale values lie beyond ± 2.75. The result of this outlier deletion is shown in Figure 2.52.

Direct manipulation provides an effective way to delete outliers. Since we identify outliers through the visualization process, it is natural to visually designate points for deletion by touching them with an input device. By their nature, outliers are typically few in number, so the process can be carried out quickly. By contrast, in computing environments that allow only static graphics, outlier deletion must be carried out by a less natural process. Scales must be read to determine cutoffs such as the ± 2.75 used for Figure 2.52, then commands must be issued to determine those observations that do not survive the cutoff, then more commands must be issued to delete them from the data given to the graphical routine, and finally the graph must be redrawn.

Outlier deletion by direct manipulation is so attractive that in 1969 Edward Fowlkes made it one of the first tools in the arsenal of direct manipulation methods discussed in Chapter 1. And outlier deletion was among the first operations of brushing, a direct manipulation tool that will be described in Chapter 3.



2.52 The outliers in the spread-standardized residuals have been removed. Direct manipulation provides a convenient environment for such point deletion.

2.10 Visualization and Probabilistic Inference

Visualization can serve as a stand-alone framework for making inferences from data. Or, it can be used to support probabilistic inferences such as hypothesis tests and confidence intervals. The fusion times provide a good example of this interplay between visualization and probabilistic inference.

The Julesz Hypothesis

Fusing a complicated random dot stereogram requires movements of the eyes to the correct viewing angle to align the two images. These movements happen very quickly in viewing a real scene; they bring focus to the scene as we shift our attention among objects at different distances. A more complex series of movements is needed to fuse a complicated random dot stereogram for the first time. But viewers can typically drastically reduce the time needed to fuse it by repeated viewing. Practice makes perfect because the eyes learn how to carry out the movements.

From informal observation, Bela Julesz noted that prior information about the visual form of an object shown in a random dot stereogram seemed to reduce the fusion time of a first look [57]. It certainly makes sense that if repeated viewing can reduce the fusion time, then seeing a 3-D model of the object would also give the viewer some information useful to the eye movements. This Julesz observation led to the experiment on fusion time, a rough pilot experiment to see if strong effects exist.

Reproducibility

Our visualization of the fusion times showed an effect; the distribution shifts toward longer times for reduced prior information. But the r-f plot showed the effect is small compared with the overall variation in fusion time. This makes it less convincing. Perhaps the better performance of increased information is spurious. Another experiment run in the same way might yield data in which decreased prior information produced shorter times. Once we begin thinking like this, it is time for probabilistic inference, not that we expect a definitive answer, but just simply to extract further information to help us judge the reproducibility of the results.

Statistical Modeling

But we must make the big intellectual leap required in probabilistic inference. We imagine that the subjects in the experiment are a random sample of individuals who can use a stereogram to see in depth. (A small fraction of individuals get no depth perception from stereo viewing.) Thus the log times for each group, NV and VV, are a random sample of the population of times for all of our specified individuals. Let μ_v be the population mean of the log VV times and let μ_n be the population mean of the log NV times.

Validating Assumptions

The normal q-q plots in our analysis showed that the two distributions of log times are reasonably well approximated by normal distributions. The lower tails are slightly elevated, but the departures are small. Thus it is reasonable to suppose that our two populations of log fusion times are well approximated by the normal distribution. The q-q plots and the fitting for viewing showed that it is reasonable to suppose the two population distributions have an additive shift. In particular, this means the standard deviations of the two populations are the same. With this checking of assumptions through visualization, we have earned the right to use the standard t-method to form a confidence interval for the difference in the two population means. Also, the visualization has revealed that it would be improper to use this method on the original data, which are severely nonnormal and do not differ by an additive shift.

Confidence Intervals

The sample mean of the 35 log VV times is $\bar{x}_v = 2.00 \log_2$ seconds. The sample mean of the 43 log NV times is $\bar{x}_n = 2.63 \log_2$ seconds. The estimate of the sample standard deviation from the residuals, $\widehat{\varepsilon}_i$, is

$$s = \sqrt{\frac{1}{76} \sum_{i=1}^{78} \widehat{\varepsilon}_i{}^2} = 1.18 \ \log_2 \ \text{seconds} .$$

A 95% confidence interval for the difference in population means, $\mu_n - \mu_v$, is

$$(\bar{x}_n - \bar{x}_v) \pm 1.99\, s \sqrt{\frac{1}{43} + \frac{1}{35}} ,$$

where 1.99 is the 0.975 quantile of a t-distribution with 76 degrees of freedom. The lower and upper values of the 95% interval are 0.09 and 1.15. Furthermore, the interval (0, 1.24) is a 97.7% interval. In other words, there is reasonable evidence that the results of the experiment are reproducible because a value of 0, which means no effect, does not enter a confidence interval until the confidence level is 97.7%.

Rote Data Analysis

The fusion-time experimenters based their conclusions on rote data analysis: probabilistic inference unguided by the insight that visualization brings. They put their data into an hypothesis test, reducing the information in the 78 measurements to one numerical value, the significance level of the test. The level was not small enough, so they concluded that the experiment did not support the hypothesis that prior information reduces fusion time. Without a full visualization, they did not discover taking logs, the additive shift, and the near normality of the distributions. Such rote analysis guarantees frequent failure. We cannot learn efficiently about nature by routinely taking the rich information in data and reducing it to a single number. Information will be lost. By contrast, visualization retains the information in the data.